

Average Treatment Effects

ECMT 676 (Econometric II) Lecture Notes

TAMU

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Introduction

- What kinds of questions can we answer using a “modern” approach to treatment effect estimation? Here are some examples:
 1. What are the effects of a job training program on employment or labor earnings?
 2. What are the effects of a school voucher program on student performance?
 3. Does a certain medical intervention increase the likelihood of survival?
- The main issue in program evaluation concerns the assignment of the binary intervention, or “treatment.”

- For example, is the “treatment” randomly assigned? (Hardly ever in business and economics, and problematical even in clinical trials because those chosen to be eligible can and do opt out.)
- A more reasonable possibility is that the treatment is effectively randomly assigned conditional on observable covariates. (“Ignorability” of treatment, “unconfoundedness,” or “selection on observables.” Sometimes called “exogenous treatment.”)

- Or, does assignment depend fundamentally on unobservables, where the dependence cannot be broken by controlling for observables? (“Nonignorable” treatment, “confounded” assignment, “selection on unobservables,” or “endogenous treatment.”)
- Often there is a component of self-selection in program evaluation.

Counterfactual Outcomes and Parameters of Interest

- Rubin Causal Model
- For each population unit, two possible outcomes:

$$\begin{cases} y_0 : \text{the outcome without treatment} \\ y_1 : \text{the outcome with treatment} \end{cases}$$

- The binary “treatment” indicator is w , where

$$w = \begin{cases} 1 : \text{treatment} \\ 0 : \text{no treatment} \end{cases}$$

- y_0 and y_1 can be discrete or continuous
- The gain from treatment (the *treatment effect*) is

$$y_1 - y_0.$$

- For a particular unit i , the gain from treatment is

$$y_{i1} - y_{i0}.$$

- Problem: For each unit i , only one of y_{i0} and y_{i1} is observed.
- In effect, we have a missing data problem.

- The **average treatment effect (ATE)** is

$$\tau_{ate} = E(y_1 - y_0).$$

The expected gain for a randomly selected unit from the population.

- The **average treatment effect on the treated (ATT)** is the average gain from treatment for those who actually were treated:

$$\tau_{att} = E(y_1 - y_0 | w = 1)$$

- With heterogeneous treatment effects ($y_{i1} - y_{i0}$ depends on i), ATE and ATT can be very different.
- Under homogeneous treatment, ATE and ATT are the same.
- The ATE might average across the gain from units that would be very unlikely to be subject to treatment.

- τ_{ate} has “external validity” in that it tells us something about a randomly drawn unit from the population. τ_{att} is specific to the particular program assignment mechanism.
- Important point: τ_{ate} and τ_{att} are defined without reference to a model or a discussion of the nature of the treatment. In particular, these definitions hold when whether assignment is randomized, unconfounded, or endogenous.
- How we estimate τ_{ate} and τ_{att} depends on what we assume about treatment assignment.
- Here we only focus on τ_{ate} (see the text for the parallel discussions on τ_{att}).

- The data we would like to have is $\{(y_{i0}, y_{i1}) : i = 1, \dots, N\}$, but we only observe w_i and

$$y_i = (1 - w_i)y_{i0} + w_i y_{i1}.$$

- Thus

$$E(y|w) = (1 - w)E(y_0|w) + wE(y_1|w)$$

Estimation under Random Assignment (RA)

- Strongest form of random assignment: (y_0, y_1) is independent of w .
- Then $E(y_0|w) = E(y_0)$ and $E(y_1|w) = E(y_1)$, and so

$$E(y|w) = (1 - w)E(y_0) + wE(y_1)$$

- Thus

$$E(y|w = 1) = E(y_1)$$

$$E(y|w = 0) = E(y_0)$$

- So τ_{ate} is identified

$$E(y|w = 1) - E(y|w = 0) = E(y_1) - E(y_0) \equiv \tau_{ate}.$$

- What is $E(y|w = 1) - E(y|w = 0)$ estimating if RA does not hold?

$$\begin{aligned} E(y|w = 1) - E(y|w = 0) &= E(y_1|w = 1) - E(y_0|w = 0) \\ &= \underbrace{E(y_1|w = 1) - E(y_0|w = 1)}_{\text{ATT}} + \underbrace{E(y_0|w = 1) - E(y_0|w = 0)}_{\text{selection bias}}. \end{aligned}$$

- Selection bias $\neq 0$ since those who go to college ($w = 1$) likely have higher value of y_0 .

- We can then estimate $E(y|w = 1)$ by the sample average on the treated subsample, and similarly for $E(y|w = 0)$.
- The estimator $\hat{\tau}_{ate}$ is just the simple difference-in-means estimator.
- The randomization of treatment needed for the simple comparison-of-means estimator to consistently estimate the ATE is rare in practice. (Eligibility is sometimes randomly assigned, but actual participation need not be.)

- Rather than assume random assignment, for each unit i we also draw a vector of covariates, $\mathbf{x}_i$.

A.1. Ignorability (Unconfoundedness, selection on observables):

$$(y_0, y_1) \perp w \mid \mathbf{x},$$

where $\perp$ means “independent”.

- w and (y_0, y_1) might be correlated but not once we control for characteristics $\mathbf{x}$.
- A useful way to express ignorability (conditional on $\mathbf{x}$):

$$D(w|y_0, y_1, \mathbf{x}) = D(w|\mathbf{x}),$$

where $D(\cdot|\cdot)$ denotes conditional distribution.

- A weaker version of ignorability (but still pretty strong) is

A.1/. Ignorability in Conditional Mean:

$$E(y_g | w, \mathbf{x}) = E(y_g | \mathbf{x}), \quad g = 0, 1. \quad ((7))$$

- From above discussion, have to think about what should be included in $\mathbf{x}$. Use the same reasoning as in multiple regression analysis in deciding whether to “hold something fixed.”

A.2. Overlap: For all $\mathbf{x}$ in its support $\mathcal{X}$,

$$0 < P(w = 1|\mathbf{x}) < 1. \quad ((8))$$

In other words, each unit in the defined population has some chance of being treated and some chance of not being treated.

- We define the **propensity score** as

$$p(\mathbf{x}) = P(w = 1|\mathbf{x}), \mathbf{x} \in \mathcal{X}. \quad ((9))$$

- **Strong Ignorability** [Rosenbaum and Rubin (1983)] = Ignorability + Overlap.

- We now turn to identification of τ_{ate} and τ_{att} .
- Under strong ignorability, we can identify the ATE conditional on $\mathbf{x}$ and therefore τ_{ate} and τ_{att} .
- In fact, for τ_{att} we can get by with a weaker version of ignorability,

$$y_0 \perp w \mid \mathbf{x}, \quad ((6'))$$

which allows w to be correlated with the (unobserved) gain, $y_1 - y_0$.

- A weaker overlap assumption suffices, too:

$$p(\mathbf{x}) < 1, \mathbf{x} \in \mathcal{X}.$$

Identification of ATEs

- Use two ways to show the treatment effects are identified under ignorability.
 - ▶ Based on regression functions
 - ▶ Based on propensity score weighting

Identification of ATEs

using regression functions

- Define the **average treatment effect conditional on $\mathbf{x}$** as

$$\tau(\mathbf{x}) = E(y_1 - y_0 | \mathbf{x}) = E(y_1 | \mathbf{x}) - E(y_0 | \mathbf{x}) \stackrel{def}{=} \mu_1(\mathbf{x}) - \mu_0(\mathbf{x}).$$

- The function $\tau(\mathbf{x})$ is of interest in its own right, as it provides the mean effect for *different segments* of the population described by the observables, $\mathbf{x}$.
- By iterated expectations,

$$\tau_{ate} = E(y_1 - y_0) = E[\tau(\mathbf{x})] = E[\mu_1(\mathbf{x}) - \mu_0(\mathbf{x})]$$

- Thus τ_{ate} is identified if $\mu_0(\cdot)$ and $\mu_1(\cdot)$ are identified.

- To see $\mu_0(\cdot)$ and $\mu_1(\cdot)$ are identified under ignorability,

$$\begin{aligned}
 E(y|\mathbf{x}, w) &= (1 - w)E(y_0|\mathbf{x}, w) + wE(y_1|\mathbf{x}, w) \\
 &\stackrel{\text{ignorability}}{=} (1 - w)E(y_0|\mathbf{x}) + wE(y_1|\mathbf{x}) \\
 &\equiv (1 - w)\mu_0(\mathbf{x}) + w\mu_1(\mathbf{x}).
 \end{aligned} \tag{1}$$

So

$$\begin{aligned}
 E(y|\mathbf{x}, w = 0) &= \mu_0(\mathbf{x}) \\
 E(y|\mathbf{x}, w = 1) &= \mu_1(\mathbf{x}).
 \end{aligned}$$

- What if we don't have ignorability?

- The functions $E(y|\mathbf{x}, w = 0)$, $E(y|\mathbf{x}, w = 1)$ are consistently estimable.
- Overlap assumption makes sense.
- By definition $E(y|\mathbf{x}, w = 0)$ will be estimated only using the control group and $E(y|\mathbf{x}, w = 1)$ will be estimated only using the treatment group.

Identification of ATEs

using propensity score weighting

- We can also establish identification using propensity score weighting.
- It turns out that

$$\tau_{ate} = E \left[\frac{wy}{p(\mathbf{x})} - \frac{(1-w)y}{1-p(\mathbf{x})} \right] = E \left\{ \frac{[w - p(\mathbf{x})]y}{p(\mathbf{x})[1 - p(\mathbf{x})]} \right\}. \quad (2)$$

- Clear from (2) that the overlap assumption is needed.
- Intuitively, if we want an average effect over the stated population, then at each $\mathbf{x}$ there must be units in the control and treatment groups.

- To show (2),

$$E \left[\frac{wy}{p(\mathbf{x})} \right] = E \left[\frac{wy_1}{p(\mathbf{x})} \right] = E \left[E \left(\frac{wy_1}{p(\mathbf{x})} \middle| \mathbf{x} \right) \right]$$

$$\stackrel{\text{ignorability}}{=} E \left[\frac{E(w|\mathbf{x})E(y_1|\mathbf{x})}{p(\mathbf{x})} \right] = E[E(y_1|\mathbf{x})] = E(y_1).$$

- Interestingly, we only need ignorability holds given $p(\mathbf{x})$, which is weaker than given $\mathbf{x}$ (more on this later).
- A similar argument shows

$$E \left[\frac{(1-w)y}{1-p(\mathbf{x})} \right] = E(y_0).$$

- Putting the two expressions together gives (2).

Estimating ATEs

- Under ignorable treatment and overlap, there are three general approaches
 - ▶ regression-based methods
 - ▶ propensity score methods
 - ▶ matching methods
- Sometimes regression or matching are done on the propensity score. We will discuss the pros and cons of such methods.

Estimating ATEs

regression-based: linear regression

Regression Adjustment

- First step is to obtain $\hat{m}_0(\mathbf{x})$ from the “control” subsample, $w_i = 0$, and $\hat{m}_1(\mathbf{x})$ from the “treated” subsample, $w_i = 1$.
- Can be as simple as (flexible) linear regression or full nonparametric regression.
- Compute fitted values in each case for *all* units in sample.
- The regression-adjustment estimates are

$$\hat{\tau}_{ate,reg} = N^{-1} \sum_{i=1}^N [\hat{m}_1(\mathbf{x}_i) - \hat{m}_0(\mathbf{x}_i)] = N^{-1} \sum_{i=1}^N \hat{\tau}(\mathbf{x}_i).$$

- If there is not sufficient overlap, $\hat{\tau}_{reg}(\mathbf{x})$ can be a poor estimator for certain values of $\mathbf{x}$.

- Now consider the case if both $m_g(\mathbf{x}) = \alpha_g + \mathbf{x}\beta_g$, for $g = 0, 1$, are linear.
- Then $\hat{m}_g(\mathbf{x}) = \hat{\alpha}_g + \mathbf{x}\hat{\beta}_g$, then

$$\begin{aligned}\hat{\tau}_{ate,reg} &= N^{-1} \sum_{i=1}^N [\hat{m}_1(\mathbf{x}_i) - \hat{m}_0(\mathbf{x}_i)] \\ &= (\hat{\alpha}_1 - \hat{\alpha}_0) + \bar{\mathbf{x}}(\hat{\beta}_1 - \hat{\beta}_0),\end{aligned}$$

where $\bar{\mathbf{x}}$ is the row vector of sample averages.

- Easiest way to obtain standard error for $\hat{\tau}_{ate,reg}$ is to use the coefficient on w_i in the regression

$$y_i \text{ on } 1, w_i, \mathbf{x}_i, w_i \cdot (\mathbf{x}_i - \bar{\mathbf{x}}), \quad i = 1, \dots, N. \quad (3)$$

$\hat{\tau}_{ate,reg}$ is the coefficient on w_i .

- Note how $\mathbf{x}_i$ is demeaned before forming interaction. This is critical because we want to estimate $\tau_{ate} = (\alpha_1 - \alpha_0) + \mu_{\mathbf{x}}(\beta_1 - \beta_0)$, not $\alpha_1 - \alpha_0$.

Why the regression (3) makes sense?

Under ignorability, we have derived (1). Thus

$$\begin{aligned} E(y|\mathbf{x}, w) &\stackrel{(1)}{=} (1-w)\mu_0(\mathbf{x}) + w\mu_1(\mathbf{x}) \\ &\stackrel{\text{linearity}}{=} (1-w)(\alpha_0 + \mathbf{x}\beta_0) + w(\alpha_1 + \mathbf{x}\beta_1) \\ &= \alpha_0 + (\alpha_1 - \alpha_0)w + \mathbf{x}\beta_0 + w\mathbf{x}(\beta_1 - \beta_0) \\ &\stackrel{\text{we want } \tau}{=} \alpha_0 + \tau_{ate}w - \mu_{\mathbf{x}}(\beta_1 - \beta_0)w + \mathbf{x}\beta_0 + w\mathbf{x}(\beta_1 - \beta_0) \\ &= \alpha_0 + \underline{\tau_{ate}w} + \mathbf{x}\beta_0 + w(\mathbf{x} - \mu_{\mathbf{x}})(\beta_1 - \beta_0), \end{aligned}$$

where $\tau_{ate} = (\alpha_1 - \alpha_0) + \mu_{\mathbf{x}}(\beta_1 - \beta_0)$ is the ATE of interest.

- The following commonly-used regression makes sense (to look at the coefficient on w_i) only when $\beta_1 = \beta_0$ (common slopes):

$$y_i \text{ on } 1, w_i, \mathbf{x}_i, \quad i = 1, \dots, N. \quad (4)$$

- One interesting case when (4) makes sense is *constant treatment effect* ($\beta_1 = \beta_0 = 0$).

Estimating ATEs

regression-based: nonlinear regression

- If linear models do not seem appropriate for $E(y_0|\mathbf{x})$ and $E(y_1|\mathbf{x})$, we can exploit the specific nature of y_g .
- If y is a binary response, estimate logit or probit separately for the $w_i = 0$ and $w_i = 1$ subsamples and average differences in predicted values:

$$\hat{\tau}_{ate,reg} = N^{-1} \sum_{i=1}^N [G(\hat{\alpha}_1 + \mathbf{x}_i \hat{\beta}_1) - G(\hat{\alpha}_0 + \mathbf{x}_i \hat{\beta}_0)]. \quad ((23))$$

- Use the same approach even under $\beta_1 = \beta_0$ or constant treatment effect.

Estimating ATEs

Propensity Score Weighting-based

- The formula that establishes identification of τ_{ate} suggests

$$\tilde{\tau}_{ate,psw} = N^{-1} \sum_{i=1}^N \frac{[w_i - p(\mathbf{x}_i)]y_i}{p(\mathbf{x}_i)[1 - p(\mathbf{x}_i)]}.$$

- $\tilde{\tau}_{ate,psw}$ is not feasible because it depends on the propensity score $p(\cdot)$.

- $p(\cdot)$ will be estimated by $\hat{p}(\cdot)$:

$$\hat{\tau}_{ate,psw} = N^{-1} \sum_{i=1}^N \frac{[w_i - \hat{p}(\mathbf{x}_i)]y_i}{\hat{p}(\mathbf{x}_i)[1 - \hat{p}(\mathbf{x}_i)]}.$$

- Two approaches:

- (1) Model $p(\cdot)$ parametrically, in a flexible way. Can show estimating the propensity score leads to a *smaller* asymptotic variance when the parametric model is correctly specified.
- (2) Use an explicit nonparametric approach.

Regression on the Propensity Score

- The motivation is that we only need ignorability given $p(\mathbf{x})$, not on $\mathbf{x}$:

$$(y_0, y_1) \perp w \mid p(\mathbf{x}),$$

which, of course, implies

$$E[y_g | p(\mathbf{x}), w] = E[y_g | p(\mathbf{x})], \quad g = 0, 1.$$

- In other words, it is sufficient to condition only on the propensity score to break the dependence between w and (y_0, y_1) . We need not condition on $\mathbf{x}$.

- By iterated expectations, $\tau_{ate} = E(y_1 - y_0) = E\{E[y_1|p(\mathbf{x})] - E[y_0|p(\mathbf{x})]\}$.
- Since

$$E[y|p(\mathbf{x}), w] = (1 - w)E[y_0|p(\mathbf{x}), w] + wE[y_1|p(\mathbf{x}), w]$$

$\stackrel{\text{ignorability on } p(\mathbf{x})}{=}$

$$(1 - w)E[y_0|p(\mathbf{x})] + wE[y_1|p(\mathbf{x})]$$

we have

$$E[y|p(\mathbf{x}), w = 0] = E[y_0|p(\mathbf{x})]$$

$$E[y|p(\mathbf{x}), w = 1] = E[y_1|p(\mathbf{x})]$$

- So, after estimating $p(\mathbf{x})$ using, say, flexible logit, we estimate $E[y|p(\mathbf{x}), w = 0]$ and $E[y|p(\mathbf{x}), w = 1]$ using the subsamples of nontreated and treated, respectively. Could use nonparametric methods.

- In the linear case, $E[y_g | p(\mathbf{x})] = \alpha_g + \gamma_1 p(\mathbf{x})$, $g = 0, 1$, and we use

$$y_i \text{ on } 1, \hat{p}(\mathbf{x}_i) \text{ for } w_i = 0 \text{ and } y_i \text{ on } 1, \hat{p}(\mathbf{x}_i) \text{ for } w_i = 1, \quad ((33))$$

which gives fitted values $\hat{\alpha}_0 + \hat{\gamma}_0 \hat{p}(\mathbf{x}_i)$ and $\hat{\alpha}_1 + \hat{\gamma}_1 \hat{p}(\mathbf{x}_i)$, respectively.

- A consistent estimator of τ_{ate} is

$$\hat{\tau}_{ate, regps} = N^{-1} \sum_{i=1}^N [(\hat{\alpha}_1 - \hat{\alpha}_0) + (\hat{\gamma}_1 - \hat{\gamma}_0) \hat{p}(\mathbf{x}_i)]. \quad ((34))$$

- Inference: Same as using usual statistics on w_i in the regression

$$y_i \text{ on } 1, w_i, \hat{p}(\mathbf{x}_i), w_i \cdot [\hat{p}(\mathbf{x}_i) - \hat{\mu}_{\hat{p}}], i = 1, \dots, N \quad ((35))$$

where $\hat{\mu}_{\hat{p}} = N^{-1} \sum_{i=1}^N \hat{p}(\mathbf{x}_i)$.

- Somewhat more common (and less desirable) is to drop the interaction term.

$$y_i \text{ on } 1, w_i, \hat{p}(\mathbf{x}_i), i = 1, \dots, N. \quad ((36))$$

- Because $0 < p(\mathbf{x}) < 1$, linearity of $E[y_g | p(\mathbf{x})]$ can be unrealistic. For a better fit, might use functions of the log-odds ratio,

$$\hat{r}_i \equiv \log \left[\frac{\hat{p}(\mathbf{x}_i)}{1 - \hat{p}(\mathbf{x}_i)} \right],$$

as regressors when y has a wide range. So, regress y_i on $1, \hat{r}_i, \hat{r}_i^2, \dots, \hat{r}_i^Q$ for some Q using both the control and treated samples, and then average the difference in fitted values to obtain $\hat{\tau}_{ate, regprop}$.

- Note that regression on the propensity score (or functions of it) require us to model $E[y_g | p(\mathbf{x})]$, while PS weighting does not.

Estimating ATEs

Matching-based

- Matching estimators are based on imputing a value on the counterfactual outcome for each unit.
- That is, for a unit i in the control group, we observe y_{i0} , but we need to impute y_{i1} . For each unit i in the treatment group, we observe y_{i1} but need to impute y_{i0} .
- For τ_{ate} , matching estimators take the general form

$$\hat{\tau}_{ate,match} = N^{-1} \sum_{i=1}^N (\hat{y}_{i1} - \hat{y}_{i0})$$

- Looks like regression adjustment but the imputed values are not fitted values from regression.
- Abadie and Imbens (2006, Econometrica) consider several approaches.

- The simplest is to find a *single match* for each observation.
- Suppose i is a treated observation ($w_i = 1$). Then

$$\begin{aligned}\hat{y}_{i1} &= y_i \\ \hat{y}_{i0} &= y_h\end{aligned}$$

for h such that $w_h = 0$ and unit h is “closest” to unit i based on some metric (distance) in the covariates.

- In other words, for the treated unit i we find the “most similar” untreated observation, and use its response as y_{i0} .
- Similarly, if $w_i = 0$, $\hat{y}_{i0} = y_i$, $\hat{y}_{i1} = y_h$ where now $w_h = 1$ and $\mathbf{x}_h$ is “closest” to $\mathbf{x}_i$.

- More generally, we can impute the missing values using an average of M nearest neighbors (possibly with ties).
- If $w_i = 1$ then

$$\begin{aligned}\hat{y}_{i1} &= y_i \\ \hat{y}_{i0} &= M^{-1} \sum_{h \in \mathfrak{N}_M(i)} y_h\end{aligned}$$

where $\mathfrak{N}_M(i)$ contains the M untreated nearest matches to observation i , based on the covariates. So for all $h \in \mathfrak{N}_M(i)$, $w_h = 0$.

- Similarly, if $w_i = 0$,

$$\begin{aligned}\hat{y}_{i0} &= y_i \\ \hat{y}_{i1} &= M^{-1} \sum_{h \in \mathfrak{S}_M(i)} y_h\end{aligned}$$

where $\mathfrak{S}_M(i)$ contains the M treated nearest matches to observation i .

- Remarkably, in the general M case can write the matching estimator as

$$\hat{\tau}_{ate,match} = N^{-1} \sum_{i=1}^N (2w_i - 1)[1 + K_M(i)]y_i,$$

where $K_M(i)$ is the number of times observation i is used as a match. (See Abadie and Imbens.)

- $K_M(i)$ is a function of the data on $(w, \mathbf{x})$, which is important for variance calculations.
- Under ignorability, $(w, \mathbf{x})$ are effectively “exogenous.”
- It is also possible to match on the estimated propensity score. This is computationally easier when the dimension in $\mathbf{x}$ is high because it is a single variable with range in $(0, 1)$.